G.C.E. (O/L) · MATHEMATICS ENGLISH MEDIUM

QUESTION PAPER

Model Paper — 06

Mathematics II

Prepared by **Chalaka Perera** · BSc (Hons) Computer Engineering (First Class) · University of Peradeniya

Time: 3 hours (+ 10 minutes reading time) **Total marks:** 100

Index number:

Important:

- Select **five** questions from Part A and **five** questions from Part B, and answer **ten** questions in all.
- Show the relevant steps and write the correct units in your answers.
- Each question carries 10 marks.
- Write your answers on separate answer sheets. Where a question says "copy the figure", copy it on your answer sheet. Use the first 10 minutes to read the paper, select the questions and decide the order in which to answer them.
- Volume of a right pyramid with base area A and height h is $\frac{1}{3}\,A\,h$. Volume of a right circular cylinder with base radius r and height h is $\pi r^2 h$.

Part A

Answer only five questions.

1. Pavithra, the owner of a supermarket, plans to buy a refrigerator priced at Rs 60 000. She has two options. **Option 1:** take a bank loan of Rs 60 000 for 10 months at a simple interest rate of 18% per year, and settle the loan and the interest in one payment at the end of the 10 months. **Option 2:** buy it on 10 equal monthly instalments of Rs 6 550 each. Under this option, Rs 6 000 of the price is repaid every month, and the interest for each month is calculated on the balance owed at the beginning of that month.

(i) Find the interest Pavithra has to pay under Option 1. (2 marks)

(ii) Find the total amount paid and the total interest paid under Option 2. (2 marks)

(iii) Show that the annual interest rate under Option 2 is higher than the annual interest rate under Option 1. (6 marks)

2. An incomplete table of values for the function $y = f(x)$ for $0 \leq x \leq 8$ is given below. The graph of the function is symmetrical about a vertical line.

x	0	1	2	3	4	5	6	7	8
y	−7	0	5	8	9	...	5	0	−7

(i) Using the symmetry of the graph, find the value of y when $x = 5$. (1 mark)

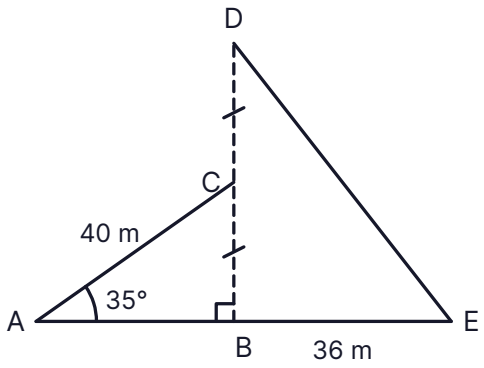
(ii) Taking 2 cm along the x-axis for 1 unit and 2 cm along the y-axis for 2 units, draw the graph of the function on graph paper. (3 marks)

(iii) (a) Write down the x-coordinates of the points where the graph meets the x-axis. (2 marks)
(b) Using these, write the function in the form $y = -(x + p)(x + q)$.

(iv) The straight line $y = 7$ meets the graph at A and B. Using the graph, (a) write the range of values of x for which $y > 7$; (b) find the length of AB to the first decimal place. (4 marks)

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3. The figure shows the cross-section of a slope in the grounds of a hospital. A, B and E are on level ground, with B between A and E. The straight path AC rises with ∠CAB = 35° and AC = 40 m. BCD is a vertical line, DC = BC, and a cable DE runs down to E, with BE = 36 m. **Copy the figure on your answer sheet and mark the given data on it.** (Use the trigonometric tables.)



(i) Find the length of BC to the nearest 0.01 m. (2 marks)

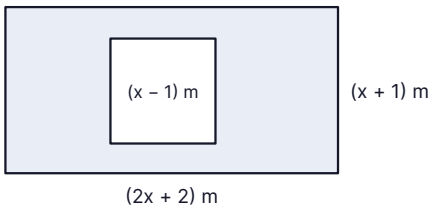
(ii) Hence find the length of BD. (1 mark)

(iii) Find the value of tan ∠DEB and hence the size of ∠DEB. (4 marks)

(iv) Show that the angle of inclination of DE is greater than the angle of inclination of AC by more than 16°. (3 marks)

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4. Asanka's paddy field is a rectangle of length $(2x + 2)$ m and width $(x + 1)$ m. A square pond of side $(x - 1)$ m has been dug inside it. The remaining shaded part of the field has an area of 100 m^2.



(i) Show that x satisfies the equation $x^2 + 6x - 99 = 0$, and solve it to find the value of x to the nearest whole number. (Take $\sqrt{3} = 1.73$.) (5 marks)

(ii) Taking $x = 7$, show that the diagonal of the pond is longer than 8 m. (Take $\sqrt{2} = 1.41$.) (5 marks)

5. Gayan bought 80 packs of detergent for his supermarket, some small packs at Rs 95 each and some large packs at Rs 150 each, for a total of Rs 9 250. Let the number of small packs be x and the number of large packs be y.

(i) Form a pair of simultaneous equations in x and y and solve them to find the number of small packs and the number of large packs. (5 marks)

(ii) Each large pack contains 150 g more detergent than each small pack, and all 80 packs together contain 24.5 kg of detergent. Find the mass of detergent in a small pack. (5 marks)

6. Thilini, a nurse at a hospital clinic, recorded the number of patients treated at the clinic on each of 40 days. The data are given in the table.

Number of patients per day	15–24	25–34	35–44	45–54	55–64
Number of days	4	9	14	9	4

(i) Find the mean number of patients treated per day. (3 marks)

(ii) Using this mean, find the expected number of patients treated in 30 days. (2 marks)

(iii) Each patient pays a fee of Rs 250, of which 40% goes to the hospital. Find the amount the hospital is expected to receive in 20 days. (3 marks)

(iv) Find the minimum possible total number of patients treated on the days on which fewer than 35 patients were treated. (2 marks)

Part B

Answer only five questions.

7. Use only a ruler with a cm/mm scale and a compass for the constructions in this question. Construction lines must be clearly visible.

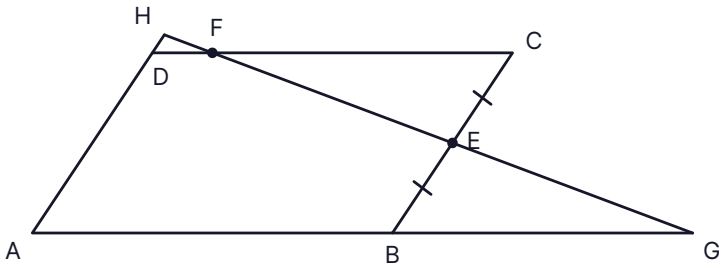
(i) Construct the line segment AB = 6.5 cm. At B, construct the perpendicular to AB, and mark on it the point C such that BC = 4.5 cm. Join AC. (3 marks)

(ii) Construct the perpendicular bisector of AC, and with AC as the diameter, draw a circle. (3 marks)

(iii) Construct the bisector of ∠ABC. Mark the point where it meets the circle again as D, and join AD and CD. (2 marks)

(iv) Giving reasons, write down the sizes of ∠ADC and ∠DAC. (2 marks)

8. ABCD is a parallelogram. E is the midpoint of BC, and F is a point on DC such that DF = $\frac{1}{6}$ DC. AB produced and FE produced meet at G, and AD produced and EF produced meet at H. **Copy the figure on your answer sheet.**



(i) Show that the triangles BGE and CFE are congruent. (3 marks)

(ii) Join BF and GC, and show that BGCF is a parallelogram. (2 marks)

(iii) Show that the triangles DFH and BGE are equiangular. (2 marks)

(iv) Hence show that DH = $\frac{1}{10}$ AD. (3 marks)

9. Asanka, who works in a hospital workshop, melts 4 identical solid metal pyramids without any loss of metal and makes one solid right circular cylinder of base radius r and height $8r$. Each pyramid has a square base of side $3a$ and a height of $2a$.

(i) Show that $a^3 = \frac{\pi}{3} r^3$. (4 marks)

(ii) Taking r = 14.5 cm and π = 3.14, find the value of a^3 using logarithms, and hence find a to the nearest whole number of cm. (3 marks)

(iii) A solid wooden cuboid has a square base of side $3a$ and a height of $2a$. A pyramid-shaped piece identical to one of the metal pyramids is cut out of the cuboid. Using the value of a^3 found in (ii), find the volume of the remaining wood. (3 marks)

10. Gayan and Asanka are going to transplant paddy seedlings in two fields of equal area. Gayan transplants 120 m^2 on the first day and, on each following day, 10 m^2 more than on the previous day. He finishes the field on the 12th day.

(i) Write down the areas, in m^2, that Gayan transplants on the first three days. (1 mark)

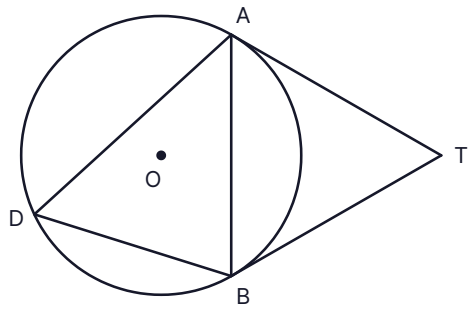
(ii) Find the area he transplants on the 12th day. (2 marks)

(iii) Find the area of the field. (2 marks)

(iv) Asanka transplants 20 m^2 more on each day than on the previous day, and finishes his field, which has the same area, on the 14th day. Find the area he transplants on the first day. (3 marks)

(v) On which day do the two of them transplant equal areas? (2 marks)

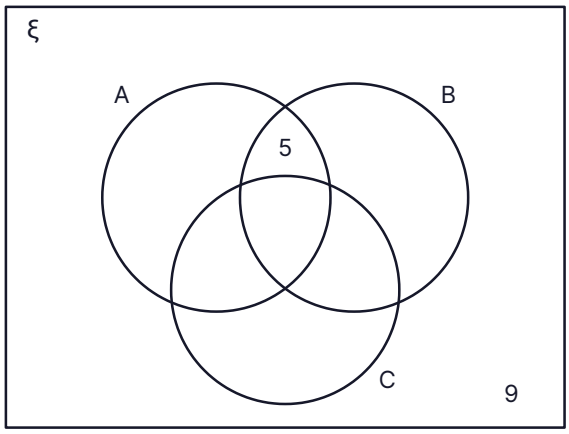
11. In the figure, O is the centre of the circle. TA and TB are the tangents drawn to the circle at A and B, and D is a point on the circle. **Copy the figure on your answer sheet and join OA, OB and OT.**



(i) Show that TAOB is a cyclic quadrilateral and that OT bisects ∠AOB. (5 marks)

(ii) (a) Write down, with a reason, an angle in triangle TAB that is equal to ∠TAB. (5 marks)
(b) Write down, with a reason, an angle at D that is equal to ∠TAB.
(c) Hence show that ∠AOT = ∠TAB.

12. Pavithra surveyed 84 farmers in a village about the crops they grew last season. Let A be the set of farmers who grew paddy, B the set who grew maize, and C the set who grew vegetables. 9 farmers grew none of the three crops, and 34 farmers grew two or more of the crops. The Venn diagram given is partly filled in. **Copy the diagram on your answer sheet and complete it as you answer the parts.**



(i) Find the number of farmers who grew exactly one of the three crops. (3 marks)

(ii) The number of farmers who grew all three crops is equal to the number who grew exactly two of the crops. Find the number who grew all three crops. (3 marks)

(iii) The number of farmers who grew only maize and vegetables is twice the number who grew only paddy and vegetables. Find each of these numbers. (2 marks)

(iv) 42 farmers grew paddy. Find the number of farmers who grew only paddy. (2 marks)

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